

Project: PoD Verification & 3PL Capacity Assessment Dashboard

(GHSC-PSM)

Background

The USAID Global Health Supply Chain Program - Procurement and Supply Management (GHSC-PSM) is committed to ensuring a continuous supply of health commodities to support U.S. Government-funded public health initiatives. To achieve its objectives, GHSC-PSM implemented last-mile distribution in ten regions, utilizing third-party logistics (3PL) transporters to mitigate the high costs associated with vehicle acquisition and fleet management by Regional Medical Stores (RMSs). To promote ongoing improvement, GHSC-PSM conducts bi-annual assessment survey on Last Mile Distribution (LMD) to assess the capacity of 3PL partners in delivering both cold chain and non-cold chain commodities, as well as to verify their delivery claims.

The dashboard was developed to:

- Monitor the performance and capacity of 3PLs across regions.
- Evaluate 3PLs capacity (fleet, temperature control, GPS, etc.).
- Track delivery timelines, cold chain compliance, and on-time delivery rates.
- Cross-check reported deliveries against receiving facility confirmations to flag discrepancies.

Approach

1. Data Collection & Sources

- 3PL Reporting Data:
 - Copies of Proof of Delivery (PoD) submitted electronically and stored on shared drive.
 - Fields: Delivery ID, Route, Commodity Type, Volume, Delivery Date, Cold Chain Requirement, Delivery Status.
- Verification & 3PL Capacity Assessment Survey Data:

- Bi-annual survey data on sampled facilities and respective 3PL collected via Kobo Toolbox. Kobo Toolbox data connected to Power BI via Kobo API.
- Fields: Facility Name, Date of Last Delivery, Commodity Received, Condition, GPS Tag, Delivery Receipt Signed. 3PL Fleet size, cold storage capacity, compliance with SOPs.

2. Data Preparation & Modelling

- Data cleaned and merged in Power Query.
- Developed a unified data model:
 - Fact tables (PoD_Assessment, Vendor_Compliance)
 - Dimension tables (VendorList, VehicleSpecs, ModeOfDelivery, LMD_Sites, LMD_Benefits, etc)
- Created calculated measures using DAX (e.g., calc_InvoiceAvail, Calc_MatchedProducts, calc_NoInvoiceCount, etc)

3. Dashboard Features

- **Cards:** Regions Visited, RMSs Visited, SDPs Visited, PoD Verified, 3PL Assessed.
- **Visuals:**
 - Interactive map showing locations of regional medical stores (RMS) and healthcare facilities (SDP) visited in the country during the PoD verification and 3PL capacity assessment.
 - Stacked bar charts showing number of PoDs verified in each region visited.
 - Table showing the percentage of SDPs having their PoDs matching the copies submitted to the central level. This visual is also used to show reasons for unavailable invoices in each region.
 - Line and clustered column chart used to compare how many verified PoDs were accompanied by invoices in each region. This visual is also used to compare how many inspected invoices matched the product items delivered in each region as well as showing the status of bin cards / inventory cards updates for each visited facility in each region comparing to the PoDs verified and invoices inspected.

- Stacked bar chart showing the 3PLs' performance on on-time delivery and as well as 3PLs' performance on loss/overage instance.
- 100% Stacked bar chart showing 3PLs' performance on temperature monitoring, vehicle specifications, staff support, and mode of delivery.
- Donut chart showing the rating of LMD benefits by healthcare facilities interviewed.
- **Interactivity:**
 - **Slicer:** Filter by Year & Month.

4. **Deployment**

- Published to Power BI Service.
- Scheduled data refresh on demand.
- Shared with M&E Team, Warehousing & Transport Leads and Technical Directors via Power BI service.

Impact

- Improved accountability by validating transporter claims using independent data.
- Enabled route-level performance monitoring, guiding contract renewals and penalties.
- Informed logistics planning by identifying cold chain capacity gaps in hard-to-reach healthcare facilities.
- Strengthened data use culture within the national supply chain coordination unit.